

Python - Positional Arguments

Positional Arguments

The **list** of **variables** declared in the parentheses at the time of defining a **function** are the **formal arguments**. And, these arguments are also known as **positional arguments**. A function may be defined with any number of formal arguments.

While calling a function –

- All the arguments are required.
- The number of actual arguments must be equal to the number of formal arguments.
- They Pick up values in the order of definition.
- The type of arguments must match.
- Names of formal and actual arguments need not be same.

Positional Arguments Examples

Let's discuss some examples of Positional arguments –

Example 1

The following example shows the use of positional argument.



```
def add(x,y):
    z = x+y
    print ("x={} y={} x+y={}".format(x,y,z))
a = 10
b = 20
add(a, b)
```

It will produce the following output –

```
x=10 y=20 x+y=30
```

Here, the add() function has two formal arguments, both are **numeric**. When integers 10 and 20 passed to it. The variable "a" takes 10 and "b" takes 20, in the order of declaration. The add() function displays the addition.

Example 2

Python also raises error when the number of arguments don't match. If you give only one argument and check the result you can see an error.



```
def add(x,y):
    z=x+y
    print (z)
a=10;
add(a)
```

The error generated will be as shown below –

```
TypeError: add() missing 1 required positional argument: 'y'
```

Example 3

Similarly, if you pass more than the number of formal arguments an error will be generated stating the same –



```
def add(x,y):
    z=x+y
    print ("x={} y={} x+y={}".format(x,y,z))
add(10, 20, 30)
```

Following is the output –

```
TypeError: add() takes 2 positional arguments but 3 were given
```

Example 4

Data type of corresponding actual and formal arguments must match. Change a to a string value and see the result.

```
def add(x,y):
    z=x+y
    print (z)
a="Hello"
b=20
add(a,b)
```

It will produce the following error –

```
z=x+y
  ~^^
TypeError: can only concatenate str (not "int") to str
```

Difference between Positional and Keyword argument

The below table explains the difference between positional and keyword argument –

Positional Argument	Keyword Argument
Only the names of arguments are used to pass data to the given function.	Keyword arguments are passed to a function in name=value form.

Arguments are passed in the order defined in function declaration.

While passing arguments, their order can be changed.

Syntax: function(param1, param2,...)

Syntax: function(param1 = value1,...)

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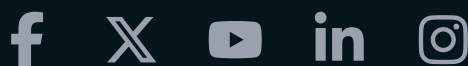
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